

GEVORG NERSESIAN

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[website](#) – [LinkedIn](#) – [GitHub](#)

EDUCATION

American University of Armenia (AUA)
Bachelor of Science in Computer Science
Capstone: Methods for Armenian Epigraphic Inscription Recognition [\[paper\]](#)

- Focused on:** theory of computation; lexical/syntax analysis, semantic checks, and code optimization; rigorous real analysis: metric spaces, convergence, functional and Fourier series; nonlinear and constrained optimization; spatial, morphological, and spectral image-processing; modeling techniques applied to stochastic and dynamical systems.
- Major GPA:** 3.46 / 4.00
- Honors:** Noubar Afeyan Foundation Grant

Yerevan, Armenia
August 2021 – June 2025

EXPERIENCE

Align Technology
Machine Learning Engineer
Machine Learning Intern

Yerevan, Armenia
June 2025 – Present
July 2024 – June 2025

- Trained CNN-based detection/classification models for anomaly detection on low-resource, high-variability datasets; delivered **>90% accuracy** using the only available samples (~100) and **cut false positives by 60%**.
- Improved a multi-stage OCR pipeline by fine-tuning ViT models and adding targeted filtering and merge-logic updates; trained on a synthetic dataset to cover unseen distributions, delivering **99.7% word accuracy** with stable performance across **6+ months** of production drift.
- Maintain and debug large production Python repositories (**300-500k+ lines** of pure code), applying solid OOP and design-pattern practices to refactor modules, fix pipeline failures, and keep ML workflows stable in daily operations.

AUA College of Science & Engineering (AUA CSE)
Junior Researcher
Student Researcher

Yerevan, Armenia
June 2025 – Present
March 2025 – May 2025

- Conduct research in 3D reconstruction of Armenian stone inscriptions to separate carved geometry from erosion and improve glyph detection on worn surfaces; in parallel developing semantic post-processing methods to infer and recover partially lost inscription text.
- Developed a CV/ML preprocessing pipeline for degraded stone inscriptions; within the same segmentation setup, performance improved from **0.64→0.92 IoU** and **0.77→0.96 Dice** (77%+ error reduction), resulting in a capstone waiver and conference acceptance of the work. Supervised by Prof. Suren Khachatryan.

PUBLICATIONS & HONORS

- Afeyan Foundation Research Grant for Armenian stone inscription OCR2025
- AUA CSE Research Showcase | placed 4th out of 602025
- Nersesian et al., Advancing Armenian Inscription Recognition. *Digital Heritage 2025*. [\[short paper\]](#)2025

CERTIFICATES

MIT's MISTI Program at AUA
Participant

Yerevan, Armenia
Seasonal, Winters 2022 - 2023

- Participated in an intensive workshop on radiation and detection methods by the leading professor of MIT in Nuclear Engineering, Prof. Areg Danagouliau; built a functioning Geiger counter using MIT's reference design. [\[course page\]](#)

Harvard University's CS50

ONLINE
Summer 2023

- A deep dive into various areas of computer science, applying in practice specific concepts, such as data structures, resource management, and security. [\[course page\]](#)

MIT's Nuclear Detection Course

Yerevan, Armenia
Summer 2022

- Continuation of an earlier workshop; worked with a new design of a Geiger Counter and enhanced it by making it more compact, adding and coding a data display; led by Prof. Areg Danagouliau. [\[course page\]](#)

SKILLS

- Languages:** Russian (native), English (fluent), Armenian (advanced)
- Coding & Data:** Python, C, R, SQL, pandas, numpy, scikit-learn
- ML & DL:** opencv, pytorch, tensorflow, CUDA, MLflow
- Tools & Platforms:** Git, DVC, AWS (EC2, S3), Linux/Windows, bash, conda